Roll No.

Total No. of Questions : 13

Total No. of Pages : 02

B.Pharma (Sem–3)

PHARMACEUTICAL ORGANIC CHEMISTRY-II

Subject Code : BP-301T

M.Code : 75105

Date of Examination : 08-06-2023

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and students have to attempt any SEVEN questions.**

SECTION-A

1. **Answer briefly :**

   a) What are the characteristic features of a compound to become aromatic?

   b) Write the nitration reaction of benzene.

   c) Draw the structure of cresol and write it's uses.

   d) Explain the acidic character of phenol with one suitable example.

   e) Define the 'Ester value'.

   f) What do you understand by Saponification of oils?

   g) Draw the chemical structure of Phenanthrene and write its medicinal uses.

   h) Write one method for the synthesis of Anthracene.

   i) Explain the greater stability of cyclohexane over cyclopentane according to Bayer's Strain Theory.

   j) What is 'Sachse Mohr's Theory'?

SECTION-B

2. Why electrophilic substitution reactions are favored on benzene? Describe the sulphonation and halogenation reactions of benzene.

3. Discuss the effect of substituents on acidity of benzoic acid and write any two reactions of benzoic acid.

4. What are fatty acids? Describe the hydrolysis and hydrogenation of fats/oils.

SECTION-C

5. Describe the resonance phenomenon in benzene and draw the structure of resonance hybrid.

6. Draw the structure of Saccharin and Chloramine and write their uses.

7. Describe the effect of substituents on basicity of amines with suitable example.

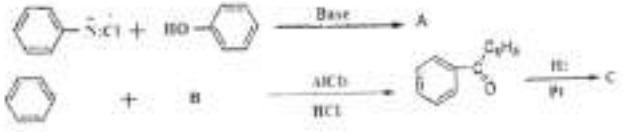
8. Describe the structural composition of Fats and oils with suitable example.

9. What is Baeyer's strain theory? Discuss the limitations of Baeyer's strain theory.

10. Define Reichert Meissl (RM) value and write its significance.

11. Write any two reactions of cyclobutane.

12. Identify A, B and C in the following reactions :



Fig

13. Describe the Friedel Craft's alkylation reaction with suitable example.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

July-2023